Vague Mass Escape Destroys Lower Semicontinuity

A counterexample for generalized transport costs on locally finite measures

FA/Banach discovery run `fa_banach_001`

`agent_lane_02` Model: `GPT5.6`

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Status: counterexample likely valid. The question in Gozlan–Herry–Peccati has a negative answer as stated, even for a bounded continuous ordinary transport cost that is linear in the conditional law.

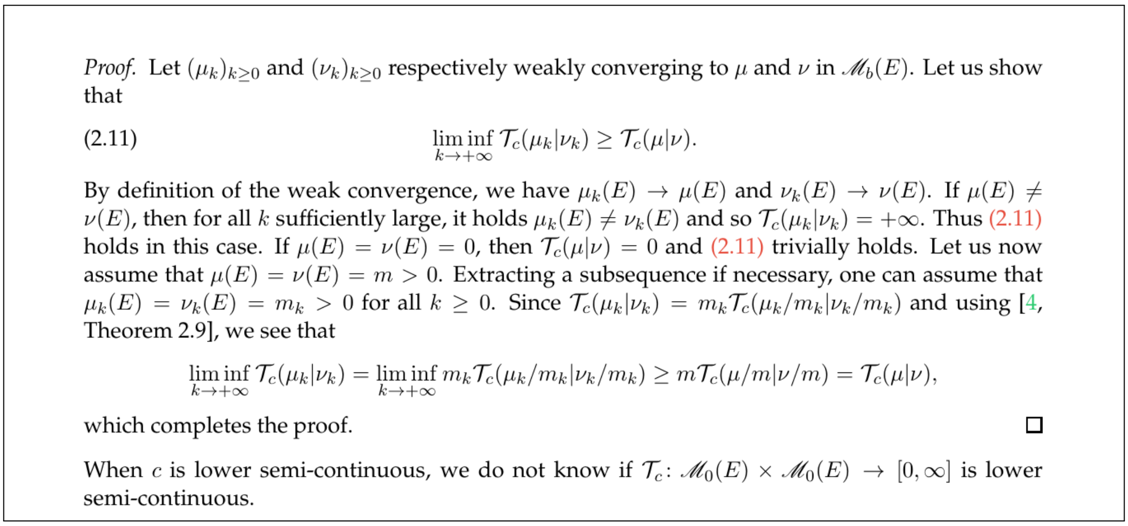
Abstract

Gozlan, Herry, and Peccati ask whether a lower-semicontinuous generalized transport cost remains lower semicontinuous after replacing finite measures with Radon measures finite on balls and weak convergence with vague convergence. We give a two-Dirac counterexample on $\mathbb{R}$. The mechanism is mass escape: vague convergence does not preserve total mass, while their extended transport functional is infinite for measures of unequal mass.

1 Source question

The source is N. Gozlan, R. Herry, and G. Peccati, *Transport inequalities for random point measures*, arXiv:2002.04923, later published in the *Journal of Functional Analysis* **281** (2021), no. 9, 109141 [1]. On page 10, at the end of Section 2.2.3, the authors ask:

> "When c is lower semi-continuous, we do not know if $\mathcal{T}_c : \mathcal{M}_0(E) \times \mathcal{M}_0(E) \to [0, \infty]$ is lower semi-continuous."



Proof. Let $(\mu_k)_{k\geq 0}$ and $(\nu_k)_{k\geq 0}$ respectively weakly converging to μ and ν in $\mathcal{M}_b(E)$. Let us show that

$$\liminf_{k\to+\infty} \mathcal{T}_c(\mu_k|\nu_k) \geq \mathcal{T}_c(\mu|\nu). \tag{2.11}$$

By definition of the weak convergence, we have $\mu_k(E) \to \mu(E)$ and $\nu_k(E) \to \nu(E)$. If $\mu(E) \neq \nu(E)$, then for all k sufficiently large, it holds $\mu_k(E) \neq \nu_k(E)$ and so $\mathcal{T}_c(\mu_k|\nu_k) = +\infty$. Thus (2.11) holds in this case. If $\mu(E) = \nu(E) = 0$, then $\mathcal{T}_c(\mu|\nu) = 0$ and (2.11) trivially holds. Let us now assume that $\mu(E) = \nu(E) = m > 0$. Extracting a subsequence if necessary, one can assume that $\mu_k(E) = \nu_k(E) = m_k > 0$ for all $k \geq 0$. Since $\mathcal{T}_c(\mu_k|\nu_k) = m_k\mathcal{T}_c(\mu_k/m_k|\nu_k/m_k)$ and using [4, Theorem 2.9], we see that

$$\liminf_{k\to+\infty} \mathcal{T}_c(\mu_k|\nu_k) = \liminf_{k\to+\infty} m_k\mathcal{T}_c(\mu_k/m_k|\nu_k/m_k) \geq m\mathcal{T}_c(\mu/m|\nu/m) = \mathcal{T}_c(\mu|\nu),$$

which completes the proof. $\square$

When c is lower semi-continuous, we do not know if $\mathcal{T}_c \colon \mathcal{M}_0(E) \times \mathcal{M}_0(E) \to [0, \infty]$ is lower semi-continuous.

Figure 1: The question in the source paper, page 10. The preceding paragraph proves lower semicontinuity for finite measures with the weak topology.

2 Definitions and convention

Let E be a Polish metric space. The source denotes by $\mathcal{M}_0(E)$ the Radon measures finite on balls and equips this space with the vague topology, generated by maps

$$\nu \longmapsto \int_E f \, d\nu$$

for continuous f vanishing outside a ball. Given $c : E \times \mathcal{P}(E) \to [0, \infty]$, it defines

$$\mathcal{T}_c(\nu_2 \mid \nu_1) = \inf_N \int_E c(x, p_x) \, \nu_1(dx), \qquad N(dx, dy) = p_x(dy)\nu_1(dx),$$

where N ranges over couplings of ν_1 and ν_2. In particular, the source explicitly adopts

$$\mathcal{T}_c(\nu_2 \mid \nu_1) = +\infty \quad \text{whenever} \quad \nu_1(E) \neq \nu_2(E). \tag{1}$$

3 Proof intuition

Weak convergence of finite measures sees the constant function 1, so total mass cannot disappear. Vague convergence does not: a unit Dirac mass may run to infinity and converge vaguely to zero. Equation (1) turns precisely this harmless-looking mass loss into an infinite jump of the transport cost. Truncating the Euclidean distance makes every prelimit cost uniformly bounded, while the limiting pair has unequal mass.

4 Counterexample

Theorem 1. *There exist a Polish space E and a bounded, jointly continuous function $c : E \times \mathcal{P}(E) \to [0, \infty)$, linear in its second argument, for which*

$$\mathcal{T}_c : \mathcal{M}_0(E) \times \mathcal{M}_0(E) \longrightarrow [0, \infty]$$

is not lower semicontinuous in the product vague topology.

Proof. Take $E = \mathbb{R}$ with its usual metric, set

$$\rho(x, y) = 1 \wedge |x - y|, \qquad c(x, p) = \int_{\mathbb{R}} \rho(x, y) \, p(dy),$$

and define, for integers $n \geq 1$,

$$\nu_{1,n} = \delta_0, \qquad \nu_{2,n} = \delta_n.$$

First, c satisfies all the advertised regularity. It is bounded by 1 and linear, hence convex, in p. If $x_j \to x$ and $p_j \to p$ weakly in $\mathcal{P}(\mathbb{R})$, then

$$\begin{aligned} |c(x_j, p_j) - c(x, p)| &\leq \sup_y |\rho(x_j, y) - \rho(x, y)| + \left| \int \rho(x, y) \, (p_j - p)(dy) \right| \\ &\leq |x_j - x| + o(1), \end{aligned}$$

because $y \mapsto \rho(x, y)$ is bounded and continuous. Thus c is jointly continuous.

The unique coupling of δ_0 and δ_n is $\delta_{(0,n)}$. Therefore

$$\mathcal{T}_c(\nu_{2,n} \mid \nu_{1,n}) = c(0, \delta_n) = 1 \qquad (n \geq 1). \tag{2}$$

On the other hand, if f is continuous and vanishes outside a ball, then $f(n) = 0$ for every sufficiently large n. Hence

$$\nu_{2,n} = \delta_n \longrightarrow 0, \qquad \nu_{1,n} = \delta_0 \longrightarrow \delta_0$$

vaguely in $\mathcal{M}_0(\mathbb{R})$. The two limiting measures have total masses 0 and 1, respectively. By (1),

$$\mathcal{T}_c(0 \mid \delta_0) = +\infty. \tag{3}$$

Combining (2) and (3) gives

$$\mathcal{T}_c(0 \mid \delta_0) = +\infty > 1 = \liminf_{n\to\infty} \mathcal{T}_c(\delta_n \mid \delta_0),$$

which is the failure of lower semicontinuity. □

5 Verification notes

1. Every measure used is Radon and finite on balls.
2. The example uses an ordinary linear transport cost, not a pathological generalized cost; c is bounded and jointly continuous.
3. Both measures have mass 1 at every prelimit index, so their coupling set is nonempty and the cost calculation is exact.
4. Only one of the two unit masses escapes every bounded set. This makes the limiting masses unequal and invokes exactly the source's $+\infty$ convention.
5. The example does not contradict the source's preceding theorem for finite measures with the weak topology. Weak convergence tests 1 and thus preserves total mass; vague convergence is the essential change.

No computation or probabilistic approximation is used. The packet's separate verification report repeats the hypothesis audit in checklist form.

6 Limitations and novelty check

This counterexample answers the question negatively in its stated generality. It does not rule out lower semicontinuity on subclasses where vague convergence is supplemented by tightness or preservation of total mass, nor does it address alternative unbalanced transport functionals that penalize rather than forbid mass loss.

A bounded search was performed on 11 August 2026. We searched the run's registry, solution, attempt, and proof-gap indexes; the exact arXiv identifier, paper title, and question phrase; and combinations of "generalized transport cost", "vague topology", "locally finite measures", "mass escape", and "lower semicontinuity", including arXiv-focused searches. The search found the original arXiv/JFA paper and general work on vague convergence, but no later paper explicitly answering this exact question. Novelty is plausible but not certified by an exhaustive bibliographic review.

7 Human-review recommendation

Recommend promotion as a full negative answer. The proof is short and exact. A reviewer should principally confirm the source convention (1) and the use of the product vague topology; both are stated in Section 2.2 of the source.

References

[1] N. Gozlan, R. Herry, and G. Peccati, *Transport inequalities for random point measures*, Journal of Functional Analysis **281** (2021), no. 9, 109141; arXiv:2002.04923.